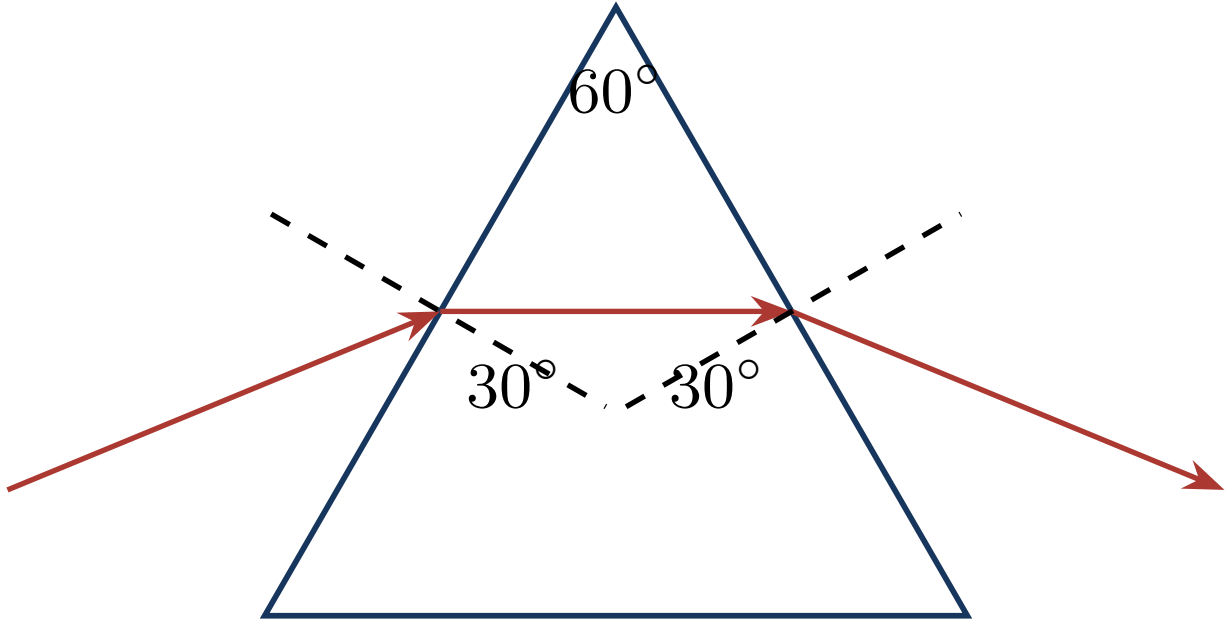

$$\theta_* = e_* \simeq 40.88^\circ, \ \alpha_{\min} \simeq 21.76^\circ$$